

KRISH BHAGIRATH

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Education

McMaster University

Bachelor of Engineering in Computer Engineering

Hamilton, ON

Sep. 2024 - 2028

- GPA: 3.9/4.0
- Awarded \$20,000+ in merit-based entrance scholarships for academic excellence.

Experience

Software Engineering Intern

May 2025 – Aug. 2025

Novologic

Toronto, ON

- Generated **10+ qualified leads** through CRM-based email campaigns, creating custom HTML/CSS templates.
- Refactored **15+** core Laravel controllers, implementing the **repository pattern** for scalable architecture.
- Optimized **SQL** and **Laravel Eloquent** queries to improve relational schema organization and query efficiency.
- Built an **AI-driven chatbot** with **React** to enhance app navigation and reduce reliance on manual support.
- Increased feature test coverage from **~65% to ~95%** through systematic edge-case testing across **agile sprints**.

Software Developer

Sep. 2024 – Present

McMaster Medical Engineering Design Team (Med-T)

Hamilton, ON

- Designed **Cradlewatch**, an **embedded infant-monitoring system** that tracks infant motion and distress using sensors and servo-controlled cameras.
- Developed embedded software integrating mechanical and electrical subsystems to ensure reliable system performance.
- Building an **NLP-based CT-scan decision model** using ClinicalBERT to identify unnecessary diagnostic imaging.

Machine Learning Developer

Oct. 2025 – Present

McMaster Aerial Robotics & Drone Club

Hamilton, ON

- Designing **real-time** circle detection algorithms with **Python & YOLO** to perform mission-specific vision tasks.
- Developing **SITL** drone simulations on a **Linux VM** with **Gazebo and DroneKit** to validate autonomy logic.
- Optimizing model inference for deployment on **Jetson Nano**, allowing for fast, stable onboard perception.

Projects

CradleWatch | *Python, Raspberry Pi, MediaPipe, OpenCV, Flask, Twilio*

- Integrated real-time **facial tracking** and **emotion detection** with MediaPipe and a Vision Transformer model.
- Deployed on **Raspberry Pi** via Picamera2 and OpenCV, enabling **servo-based camera control** and video feed.
- Implemented SMS notifications using **Twilio API** to alert caregivers of prolonged distress, delivering a fully functional prototype for proactive infant care.

DeepFakeDetector | *Python, PyTorch, CNN, Vision Transformer, FFT, Next.js, FastAPI*

- Developing a **hybrid detection model** integrating CNNs, Vision Transformers, and FFT models for robust detection of AI-generated images.
- Boosting model accuracy through advanced fine-tuning, frequency methods, and artifact-focused augmentations.
- Implementing a **Next.js and FastAPI** web app, utilizing Grad-CAM for visual heatmaps of deepfake predictions.

CourtML | *Python, React, scikit-learn, GitHub Actions, Pandas, NumPy*

- Built an **end-to-end ML pipeline** using tree-based and regression models on 8 years of NBA data, improving prediction accuracy by **~20%**.
- Improved model robustness using rolling-window feature engineering and automated evaluation on new data.
- Created a **React UI** displaying weekly NBA predictions and historical accuracy using live NBA API data.

Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript, PHP, SQL, Verilog, Linux (Ubuntu)

Frameworks & Libraries: React, Laravel, Scikit-learn, Numpy, Pandas, PyTorch, OpenCV

Other: Git, Visual Studio Code, Postman, Arduino, Raspberry Pi, FPGA, AI/ML, Cloud Infrastructure